PROJECT PROPOSAL

BioLearn BD: An Interactive Biology Learning Platform for HSC and University Admission Students

CSE412: Software Engineering
Section-6
Group No - 1

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1. Purpose

BioLearn BD is a proposed web-based learning platform designed primarily for Bangladeshi intermediate-level students preparing for the HSC examination and university admission examinations. Biology students often need to study textbook chapters, understand individual topics, review diagrams and notes, watch explanatory lessons, and practice questions from different sources. The proposed system aims to bring these learning resources together in a structured, topic-based platform so that students can study Biology in a more organized and interactive way.

The system will cover Biology 1st Paper and Biology 2nd Paper. Students will be able to select a paper, chapter, and topic, then access topic-specific notes, relevant images or diagrams, an embedded YouTube lesson, and a quick quiz. An administrator will manage chapters/topics and maintain the learning materials, video links, and quiz questions. Students can also take a chapter-end quiz after completing a chapter, track their chapter-wise progress, and compare their performance with other learners through a points-based leaderboard.

The main beneficiaries are HSC students, university admission candidates, and administrators/content managers. The system is intended to make Biology learning easier to navigate, more engaging, and more useful for revision and self-assessment.

2. Inception

The inception phase identifies the stakeholders affected by the proposed system and their expectations. For BioLearn BD, the main client-side stakeholders are students and the administrator/content manager. The development team is also a project stakeholder because it is responsible for understanding and implementing the agreed requirements.

2.1 Identifying Stakeholders

Stakeholder	Why they are a stakeholder	Role / Interaction	Expected from the system
HSC Students	They are the primary users and beneficiaries.	Browse papers, chapters and topics; study materials; watch videos; take quizzes.	Clear navigation, understandable notes, useful images, relevant videos, quick assessment and results, visible progress, and motivation to study consistently.
University Admission Candidates	They use Biology content for admission-exam preparation and revision.	Review topic-based materials and practice quick quizzes.	Fast revision, organized content, relevant practice questions and easy access to learning

			resources.
Administrator / Content Manager	The learning content must be maintained through an administrative side.	Manage chapters/topics; upload notes/images; add YouTube links; create and manage quiz questions.	Easy content management, reliable updates and control over learning materials.
Project Development Team	They design and develop the proposed system based on stakeholder requirements.	Analyze requirements, design, implement and test the system.	Clear, agreed and prioritized requirements.

2.2 Identifying Multiple Viewpoints of the Stakeholders

Student viewpoint

- Students want all Biology learning resources organized according to the existing 1st Paper and 2nd Paper structure.
- They consider topic-wise notes, relevant images/diagrams, embedded video lessons and quick quizzes important.
- They need simple navigation: Paper → Chapter → Topic → Learning Materials → Quiz.
- They may have limited time for preparation, so the system should support quick revision and self-assessment.
- Content should be readable, relevant and easy to access.
- Students need a progress-tracking section to monitor completed topics, quiz attempts, and overall learning progress.
- Students want to test their understanding after finishing a whole chapter, not only individual topics.
- They want to see how many chapters they have completed and how much of each paper remains.
- Many students are motivated by seeing their rank among other learners and want to compare their performance.

Administrator viewpoint

- The administrator wants to maintain the learning content without changing the overall system structure.
- The administrator needs to add or update notes, images, YouTube links and quiz questions for topics.
- The administrator needs control over quiz questions and correct answers.
- The administrator needs to create and manage chapter-end quizzes in addition to topic quizzes.
- The system should reduce unnecessary complexity in content management.

2.3 Working Towards Collaboration

Common Requirements:

- Organized Biology content based on 1st Paper and 2nd Paper.
- Chapter and topic-based navigation.
- Reliable and understandable learning materials.
- Quick access to relevant educational videos.
- Topic-based quizzes for self-assessment.
- Chapter-end quizzes and visible progress tracking.
- Simple and user-friendly interface.

Conflicting Requirements:

- Students may prefer a large amount of learning material, while administrators need to keep content focused and manageable.
- Students may want quick access to many resources, while the system should avoid an unnecessarily complicated interface.
- Students may prefer different learning styles, while the platform must provide a consistent structure for all topics.
- Students may be motivated by competitive rankings, but the leaderboard should not discourage students who are lower ranked.

Final Requirements:

- Provide Biology 1st Paper and 2nd Paper with their chapters and topics.
- Allow students to access topic-wise notes and relevant images/diagrams.
- Provide an embedded YouTube video for each supported topic.
- Provide a quick quiz for each supported topic.
- Provide a chapter-end quiz after each chapter to assess students' overall understanding of that chapter.
- Provide a progress tracking dashboard showing completed chapters, completed topics, quiz performance, and overall learning progress.
- Provide a leaderboard that ranks students by accumulated points, based on quiz scores and completed chapters.
- Provide an administrative interface for maintaining learning materials and quizzes.
- Allow the administrator to create and manage chapter-end quiz questions.
- Keep the navigation simple and consistent.

3. Elicitation

The elicitation phase combines stakeholder needs into requirements for the proposed system. The requirements below are organized using Quality Function Deployment (QFD) into

Normal Requirements, Expected Requirements, and Exciting Features, following the categories specified by the project requirements document.

3.1 Quality Function Deployment

The stakeholder needs are translated into system requirements according to their importance to the proposed learning platform.

3.1.1 Normal Requirements

These are the main requirements and objectives explicitly expected from the system:

- Student registration/login and role-based access to the student area.
- Admin login and access to the administrative area.
- Access to Biology 1st Paper and Biology 2nd Paper.
- Chapter and topic-based organization of Biology content.
- Topic-wise notes and learning materials.
- Relevant images and diagrams for topics.
- YouTube video link management and embedded video playback.
- Topic-based quick quiz/exam.
- Quiz questions with correct answers and marks.
- Quiz result/score display to the student.
- Chapter-end quiz after each chapter, with automatic result calculation.
- Admin management of notes, images, video links, topic quiz questions and chapter-end quiz questions.
- Chapter completion tracking.
- Student progress dashboard showing completed chapters, completed topics, quiz scores, and overall completion percentage.
- Point-based scoring system for quizzes and completed chapters.
- Leaderboard ranking students by total points.
- Display of the student’s own rank among other learners.

3.1.2 Expected Requirements

These requirements may be implicit, but their presence is important for user satisfaction:

- Simple and consistent navigation between papers, chapters and topics.
- Readable presentation of notes and learning materials.
- Images/diagrams should appear with the relevant topic rather than being separated from the lesson.
- Embedded videos should be accessible without forcing the student to leave the learning page.
- Quiz questions should be associated with the correct topic.
- Students should receive clear feedback after completing a quiz.
- Only authorized administrators should be able to modify learning content.
- The system should avoid unnecessary complexity for students.

- Progress information should be presented clearly so students can understand which topics they have completed and which still require study.
- The leaderboard should update promptly after a student completes a quiz.
- The leaderboard should display only a username or display name, not private student information.
- Points and rankings should be calculated consistently and fairly for all students.
- Students should be able to clearly see which chapters are completed, in progress, or not started.

3.1.3 Exciting Features

The following features can provide additional value beyond the basic learning requirements:

- Topic page combining notes, diagrams/images, video and quiz in one learning flow.
- Quick topic-based assessment for immediate self-evaluation.
- Search or quick navigation for finding a chapter/topic more easily.
- Progress-oriented presentation showing completed or attempted topics.
- Admission-exam-oriented practice questions as an extension of the topic quizzes.
- Visual progress tracking, such as progress bars per chapter and paper, that helps students monitor their overall learning progress.
- Competitive leaderboard that encourages students to study consistently and improve their ranking.
- Achievement badges and milestone rewards for completing chapters and scoring well in quizzes.

4. Scenario Based Modeling

4.1 Usage Scenario

Scenario 1: Student studies a topic

1. The student logs into the platform.
2. The student selects Biology 1st Paper or 2nd Paper.
3. The student selects a chapter.
4. The student selects a topic.
5. The system displays the topic notes and relevant images/diagrams.
6. The system displays an embedded YouTube lesson for the topic.
7. The student studies the material and may watch the video from the same page.
8. The student can proceed to the quick quiz.
9. The system records the student's learning activity so the student can view topic and quiz progress in the progress section.

Scenario 2: Student takes a quick quiz

1. The student opens a topic.
2. The student selects the quick quiz option.
3. The system displays the quiz questions.

4. The student submits answers.
5. The system checks the answers against the stored correct answers.
6. The system calculates the result.
7. The student views the score/result and can review the topic again.

Scenario 3: Administrator adds learning content

1. The administrator logs into the admin side.
2. The administrator selects a paper, chapter and topic.
3. The administrator adds or updates the topic note.
4. The administrator uploads or associates relevant images/diagrams.
5. The administrator adds the relevant YouTube video link.
6. The administrator saves the content.
7. The updated learning materials become available to students.

Scenario 4: Administrator creates a topic quiz

1. The administrator selects a paper, chapter and topic.
2. The administrator creates quiz questions.
3. The administrator specifies the correct answer and marks where applicable.
4. The administrator saves the quiz.
5. The quiz becomes available to students under the selected topic.

Scenario 5: Student takes a chapter-end quiz

1. The student completes the topics of a chapter.
2. The student selects the chapter-end quiz.
3. The system displays the quiz questions.
4. The student submits the answers.
5. The system checks the answers, calculates the score, and awards points.
6. The system marks the chapter as completed and updates the student's progress.

Scenario 6: Student views progress and leaderboard

1. The student logs into the platform and opens the Progress Dashboard.
2. The system displays completed chapters, completed topics, quiz scores, earned points, and overall completion percentage.
3. The student opens the Leaderboard section.
4. The system displays students ranked in descending order of total points and highlights the student's own rank.
5. The student compares their performance with other learners and identifies areas for improvement.

Scenario 7: Administrator creates a chapter-end quiz

1. The administrator selects a paper and chapter.
2. The administrator creates chapter-end quiz questions and specifies the correct answers and marks.
3. The administrator saves the quiz.
4. The quiz becomes available to students after they finish the chapter.

5. Conclusion

BioLearn BD is proposed as a structured and interactive Biology learning platform for Bangladeshi intermediate students preparing for the HSC examination and university admission examinations. By organizing Biology 1st Paper and Biology 2nd Paper into chapters and topics, the system will bring notes, relevant images or diagrams, embedded YouTube lessons, quick topic-based quizzes, and student progress tracking together in one place. The student side will provide an accessible learning, practice, and self-assessment experience, including chapter-end quizzes, a progress dashboard for monitoring completed chapters and topics, and a points-based leaderboard that motivates students to learn consistently. The administrator side will allow authorized management of learning materials, videos, images, and quiz content. Overall, the proposed system aims to make Biology learning more organized, accessible, interactive, and trackable for its target students.